Preprocessing: import modules, read images, and create the output directory for pictures.

```python
import cv2
from PIL import Image
import numpy as np
import os
import math

# read image
path = r"C:\ntu_csie\1141\cv\lena.bmp"
img = cv2.imread(path, cv2.IMREAD_COLOR)
h, w, ch = img.shape

# create output dir
output_path = r"C:\ntu_csie\1141\cv\output\hw1"
os.makedirs(output_path, exist_ok=True)
```

Part 1-a: Write the rows of the image in reverse order, from bottom to top.

```python
a = np.zeros((h, w, ch), dtype=np.uint8)
for i in range(h):
    a[i] = img[h-1-i]
cv2.imwrite(os.path.join(output_path, "part1_a.bmp"), a)
```

Part 1-b: Write the columns of the image in reverse order, from right to left.

```
b = np.zeros((h, w, ch), dtype=np.uint8)
for j in range(w):
    b[:, j] = img[:, w-1-j]
cv2.imwrite(os.path.join(output_path, "part1_b.bmp"), b)
```

Part 1-c: Write the pixels of the image with the row and column indices swapped.

```
c = np.zeros((w, h, ch), dtype=np.uint8)
for i in range(h):
    for j in range(w):
        c[j][i] = img[i][j]
cv2.imwrite(os.path.join(output_path, "part1_c.bmp"), c)
```

Part 2-d: Use the rotate function from PIL.Image to perform the rotation.

```
# (d) rotate lena.bmp 45 degrees clockwise
d = Image.open(path)
rotated = d.rotate(-45, expand=False)
rotated.save(os.path.join(output_path, "part2_d.bmp"))
```

Part 2-e: Compute the mean value of the RGB channels for each 2×2 block of the original image, and write it to an array with half the width and half the height of the original image.

```
h2, w2 = h//2, w//2
e = np.zeros((h2, w2, ch), dtype=np.uint8)
for i in range(h2):
    for j in range(w2):
        patc = img[i*2:i*2+1, j*2:j*2+1, :]
        e[i][j] = np.mean(patc, axis=(0, 1))
cv2.imwrite(os.path.join(output_path, "part2_e.bmp"), e)
```

Part 2-f: For each pixel, if the mean of the RGB values exceeds the threshold of 128, write 255 to the pixel; otherwise, write 0.

```
f = np.zeros((h, w), dtype=np.uint8)
for i in range(h):
    for j in range(w):
        r, g, b = img[i][j].astype(int)
        f[i][j] = 0 if (r+g+b)//3 < 128 else 255
cv2.imwrite(os.path.join(output_path, "part2_f.bmp"), f)
```